ANDROID APP DEVELOPMENT · TASK 3

Offline Employee Manager

Jetpack architecture components, MVVM, Room database, ViewModel, LiveData, Repository pattern and offline persistence

7	17	16	M3
screens	tests passing	Kotlin files	Material Design

DEVELOPER
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PROGRAMME
Sumerix Global · Android App Development · 36 weeks

BATCH START
15 October 2026

STACK
Kotlin 2.0 · AGP 8.7 · Material 3 · minSdk 26 / target 35

1 · Overview

Jetpack architecture components, MVVM, Room database, ViewModel, LiveData, Repository pattern and offline persistence. Package `com.eswar.employeemanager`.

- MVVM: Fragments observe `EmployeeViewModel` (LiveData + `switchMap` filters) → `EmployeeRepository` → Room DAOs
- Room database with two entities (departments 1 : * employees), foreign key, unique e-mail index, `@Relation`, GROUP BY statistics, TypeConverters and exported schema JSON
- Full CRUD offline: add / edit with field validation, swipe-to-delete with Undo, search, department filter and sort menu
- Seeded on first launch (6 departments, 12 employees) through a `RoomDatabase.Callback`
- Statistics screen (head-count and payroll by department) and CSV export through the share sheet
- One-shot UI events (`Event` wrapper) for Snackbars and navigation; dark theme and tablet grid

7
screens captured

17
automated tests · 0 failing

16
Kotlin source files

9
XML layouts

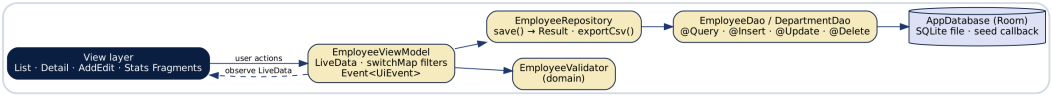
2 · Brief coverage

Brief part	Implementation
Jetpack architecture	ViewModel, LiveData, Room, Navigation, lifecycle-aware observers
MVVM	ui/EmployeeViewModel.kt, Fragments in ui/list, ui/detail, ui/edit, ui/stats
Room / SQLite	data/Entities.kt, EmployeeDao.kt, AppDatabase.kt, Converters.kt, app/schemas/.../1.json
Repository pattern	data/EmployeeRepository.kt — single source of truth, Result-based save, duplicate-email handling
Data persistence & offline	Everything runs from the local database; no network permission
Validation	domain/EmployeeForm.kt — EmployeeValidator with unit tests
Testing	DAO tests on an in-memory database, ViewModel and validator tests, screenshot tests

3 · Architecture

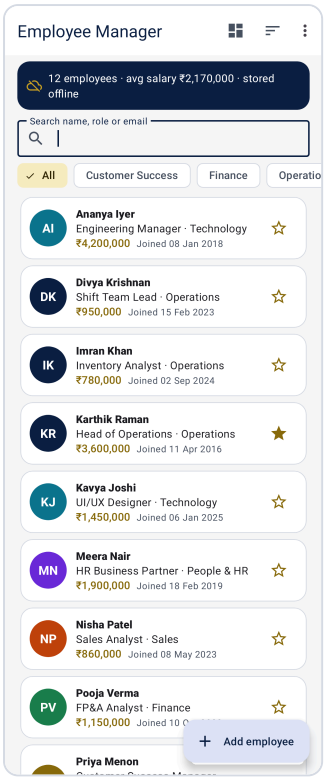


Database er

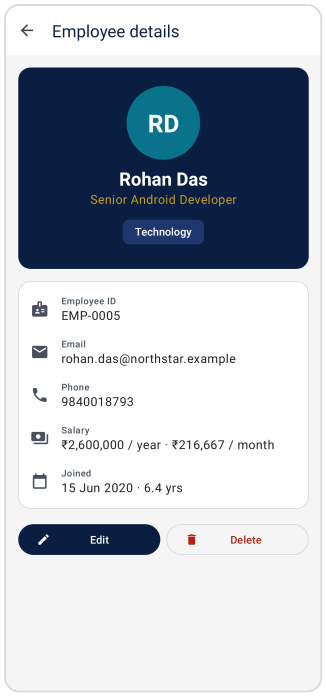


Mvvm architecture

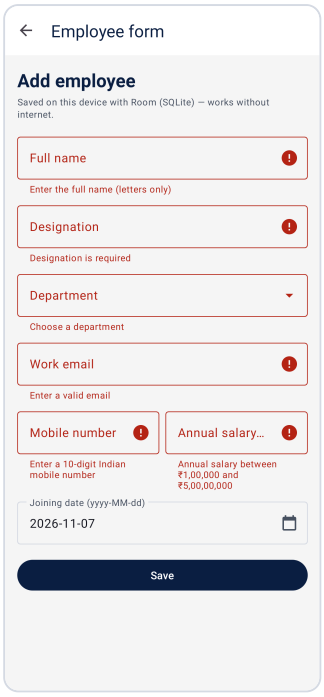
4 · Screens (phone, light & dark)



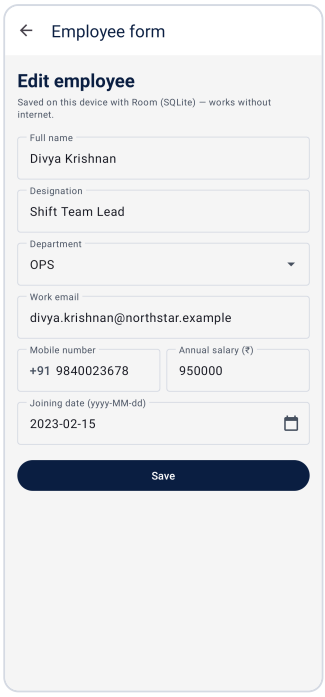
Employee list



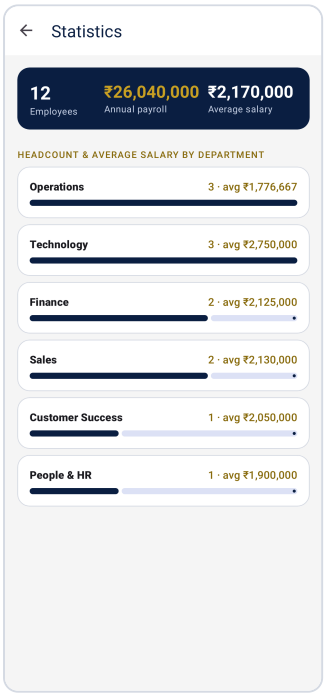
Employee detail



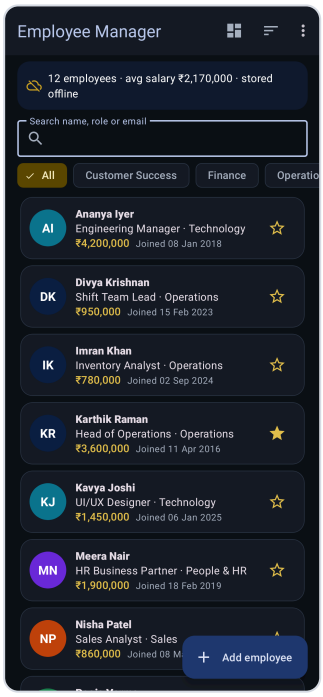
Add employee validation



Edit employee

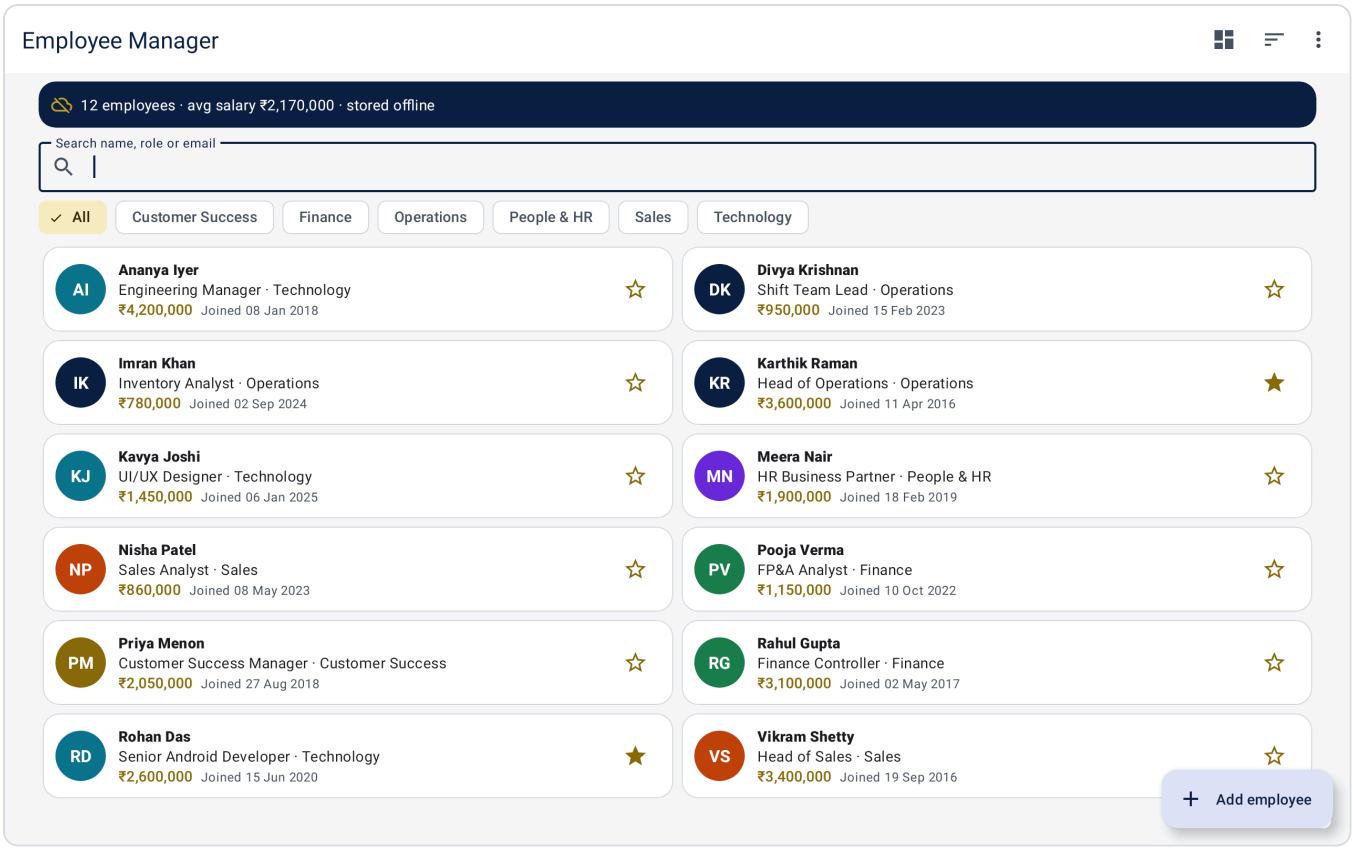


Statistics



Employee list dark

5 · Tablet layout (sw600dp)



Tablet two column

6 · Testing

Test class	Tests	Failures	Cases
EmployeeDaoTest	4	0	searchFiltersAndSorts, duplicateEmailIsRejectedNotCrashing, insertReadUpdateDelete, relationAndAggregates
EmployeeValidatorTest	3	0	everyFieldIsChecked, futureJoiningDateRejected, validFormIsNormalised
EmployeeViewModelTest	3	0	invalidFormExposesErrorsAndDoesNotSave, filterChangesRequeryRoom, validFormIsInsertedThenDeletedAndRestored
ScreenshotTest	7	0	s05_stats, s03_add_validation, s01_list, s04_edit, s06_dark, s02_detail, s07_tablet

Tests run on the JVM with Robolectric (API 34). Screenshot tests render every screen with Roborazzi in native graphics mode, so the images in this document are produced by the real XML layouts, themes and resources.

7 · Project structure

```
src/
  main/
    AndroidManifest.xml
    java/
      com/
        eswar/
          employeemanager/
            EmployeeManagerApp.kt
            data/
              AppDatabase.kt
              Converters.kt
              EmployeeDao.kt
              EmployeeRepository.kt
              Entities.kt
              SeedData.kt
            domain/
              EmployeeForm.kt
            ui/
              EmployeeViewModel.kt
              Format.kt
              MainActivity.kt
              detail/
                EmployeeDetailFragment.kt
              edit/
                AddEditEmployeeFragment.kt
              list/
                EmployeeAdapter.kt
                EmployeeListFragment.kt
              stats/
                StatsFragment.kt
    res/
      drawable/
        bg_avatar_circle.xml
        ic_add.xml
        ic_apartment.xml
        ic_arrow_up.xml
        ic_badge.xml
        ic_calendar.xml
        ic_chevron.xml
        ic_cloud_off.xml
        ic_dashboard.xml
        ic_delete.xml
        ic_edit.xml
        ic_info.xml
        ic_launcher_foreground.xml
        ic_location.xml
        ic_mail.xml
        ic_payments.xml
        ic_people.xml
        ic_phone.xml
        ic_search.xml
        ic_sort.xml
        ic_star.xml
        ic_star_border.xml
      layout/
        activity_main.xml
        fragment_add_edit.xml
        fragment_detail.xml
        fragment_employee_list.xml
        fragment_stats.xml
        item_employee.xml
        item_info_row.xml
```

```
        item_stat_row.xml
        view_filter_chip.xml
      menu/
        menu_list.xml
      mipmap-anydpi-v26/
        ic_launcher.xml
      navigation/
        nav_graph.xml
      values/
        colors.xml
        dimens.xml
        ic_launcher_background.xml
        strings.xml
        themes.xml
      values-night/
        colors.xml
      values-sw600dp/
        dimens.xml
  test/
    java/
      com/
        eswar/
          employeemanager/
            data/
              EmployeeDaoTest.kt
            domain/
              EmployeeValidatorTest.kt
            ui/
              EmployeeViewModelTest.kt
              ScreenshotTest.kt
    resources/
```